

Efficient Maximum Entropy Reinforcement Learning Framework Guided by Expert Policies: An Application in Autonomous Driving

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Abstract—Deep Reinforcement Learning (DRL) has garnered significant attention for its superior adaptability and continuous learning capabilities in the decision control of autonomous driving. However, in complex human-machine hybrid scenarios with high-dimensional continuous state spaces, such as unsignalized intersections, DRL-based autonomous driving decision-making control algorithms exhibit low sample efficiency and struggle to converge. Moreover, designing a reward function that accurately captures the complexities of human-machine interactions is highly challenging; inappropriate reward design can lead to unintended behaviors by the agent, thereby threatening system safety. To address these issues, this study proposes a Contrastive Expert-Guided Maximum Entropy Reinforcement Learning (CE-MERL) framework. This framework trains a discriminator model using both human expert datasets and the interaction experiences of the agent with the environment to identify expert policies. It then constrains the agent's policy alignment with the expert policy as a cost within the maximum entropy reinforcement learning framework. This not only enables the agent to conduct reasonable and efficient exploration, thereby improving sample efficiency, but also allows it to learn safer, human-like policies even when using only sparse rewards. To ensure that the discriminator can continuously and effectively identify expert policies, we adopt the Vision Transformer (ViT) model as a feature encoder and introduce auxiliary tasks of contrastive learning. This method achieves efficient coupling of state and action features and mitigates overfitting caused by limited expert demonstration data. Simulation results show that in two tasks: unprotected left turn and ramp merging, our proposed CE-MERL method, without the use of shaped rewards, exhibits higher task success rates and sample efficiency compared to baseline algorithms.

Index Terms—Deep reinforcement learning (DRL), human driving priority, contrastive learning, autonomous driving.

I. INTRODUCTION

IN RECENT years, with the development of deep learning technologies, autonomous vehicles (AVs) have achieved significant accomplishments in improving transportation efficiency and safety [1]. However, in complex human-machine mixed traffic environments, particularly in scenarios involving unsignalized intersections and ramp merging tasks, the

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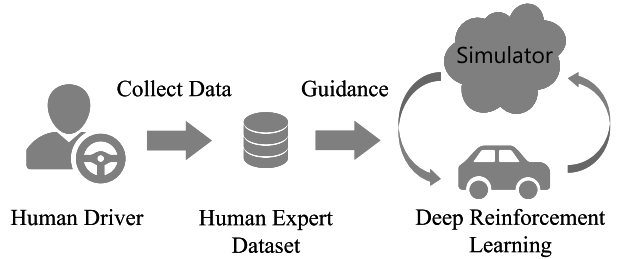


Fig. 1. The proposed CE-MERL method is a DRL approach with human guidance. It leverages human expert driving data to direct the DRL agent, thereby enhancing sample efficiency and enabling the agent to acquire safer and more robust human-like driving strategies.

decision-making reliability of AVs remains a significant challenge. DRL, as an advanced learning paradigm that leverages data-driven approaches along with its robust learning and efficient inference capabilities, has the potential to overcome this challenge [2]. However, when employing DRL to address tasks such as autonomous driving, which involves high-dimensional continuous state and action spaces, agents often fail to conduct rational and controllable exploration. This leads to the issue of low sample efficiency—i.e., the difficulty in obtaining high-quality, instructive data samples. Furthermore, designing a reward function that precisely delineates the desired behavior presents another significant challenge. Imperfect reward design may mislead AVs, giving rise to unintended behaviors, such as sudden acceleration to force other vehicles to stop and yield space, which greatly compromises the safety of autonomous vehicles.

To address the challenges of low sample efficiency and the complexity of designing reward functions in DRL for autonomous driving tasks, this paper introduces a novel Contrastive Expert-Guided Maximum Entropy Reinforcement Learning (CE-MERL) framework. CE-MERL trains a binary classifier expert policy discriminator using datasets of human expert demonstrations and interaction data stored in the replay buffer from DRL, which determines whether given scene states and action inputs conform to the distribution of expert policies. Concurrently, by utilizing the output of the expert policy discriminator, behavioral constraints are constructed and added to Maximum Entropy Reinforcement Learning (MaxEnt RL) [3], ensuring that the agent maintains a moderate exploration capability while limiting the extent of deviation from expert policies. This achieves more stable and efficient exploration, thereby enhancing sample efficiency. Additionally, the output

of the expert policy discriminator can serve as an extra reward signal, guiding the agent to select actions more consistent with expert policies under given scene conditions, thus avoiding the reliance of DRL on complex reward engineering.

In the proposed CE-MERL, the agent is required to generate experience samples that conform to the distribution of expert policies, while the task of the expert policy discriminator is to determine whether given experience samples originate from the genuine expert policy distribution. This process inherently embodies the characteristics of Generative Adversarial Networks (GAN) [4]. However, compared with the interaction data generated by the agent, the amount of demonstration data provided by human experts is typically very limited, leading to an asymmetry in data distribution. This can cause the discriminator model to overfit, capturing complex and non-critical patterns in the observation space rather than truly reflecting important features of the decision-making process. Moreover, the features provided by scene states are far richer than those provided by actions, and we desire the discriminator to discern the implicit behavioral policies in both scene states and actions, rather than comparing scene states. Therefore, given the strong capability of ViT to process multimodal input [5], [6], we made minor modifications to ViT [7] to serve as the feature encoder of an expert policy discriminator. Specifically, action embeddings are utilized as classification query tokens, thereby facilitating the coupling of scene representation with action command information. Additionally, we introduce a contrastive learning auxiliary task into the binary classification task of the discriminator, maximizing the InfoNCE mutual information [8] between state-action joint embeddings in two independently augmented views of scene states derived from expert data, and maximizing the feature discrepancy between expert data and agent interaction data. These measures not only focus the scene representation primarily on features related to action instructions but also enhance the intra-class compactness and inter-class separability of the feature representations. As a result, human expert demonstration data can continuously exert a positive influence on the agent's policy, thereby improving the overall performance of the algorithm.

In summary, the main contributions of this paper are as follows:

- We evaluate the proposed CE-MERL framework on two classic autonomous driving decision control challenges: unprotected left turns and ramp merging. The experimental results show that, compared to baseline algorithms, CE-MERL not only achieves higher task success rates but also significantly improves sample efficiency. Furthermore, through detailed experimental analysis, we validate the critical role of the ViT state-action encoder and the contrastive loss term in optimizing model performance, demonstrating the effectiveness of these methods in enhancing feature learning quality and decision-making capabilities.
- This study proposes an innovative DRL framework that integrates limited human expert demonstrations into Max-Ent RL. This method ensures the agent's exploration capability while implicitly constraining its policy distance from the expert policies, thereby significantly improv-

ing sample efficiency and avoiding reliance on complex reward engineering.

- In this study, we introduce ViT as the state-action encoder for the expert policy discriminator and incorporate a contrastive learning auxiliary task into the discriminator's learning process. This approach not only addresses the issue of feature decoupling caused by the separation of action embeddings from the perception module but also mitigates the risk of overfitting due to the limited amount of expert demonstration data.

The remainder of this paper is organized as follows: Chapter 2 reviews and analyzes recent DRL-based autonomous driving algorithms from three perspectives; Chapter 3 describes the autonomous driving scenarios involved in this study and the MaxEnt RL decision control mechanism; Chapter 4 details the implementation procedures of our method; Chapter 5 introduces the experimental setup, and the experimental results and discussions are presented and analyzed in Chapter 6. Finally, the paper concludes with a review and summary of the entire work and suggestions for future research directions.

II. RELATED WORK

A. Reinforcement Learning-Based Autonomous Driving

DRL has garnered substantial attention for its impressive human-like learning capabilities and the high-fidelity execution of inference, as emphasized in the work by Wu et al. [9] and Huang et al. [10]. Notably, Ma et al. [11] employed a safety-oriented energy function to learn policies that avoid constraint violations, thereby enabling autonomous vehicles to identify risky actions before they are taken, enhancing the safety of DRL. Other studies have introduced ViT models into DRL methods [12], using physical target states as inputs to the scene encoder to couple scene representation with target information, achieving efficient autonomous navigation. Furthermore, Liu et al. [13] combined DRL with Model Predictive Control (MPC), demonstrating the versatility and flexibility of the reinforcement learning paradigm.

However, it is important to note that model-free DRL relies on extensive exploration and iterative trial-and-error, facing challenges related to low learning efficiency and complex reward function design. In response to these issues, this study enhances the learning efficiency of DRL and enables agents to learn safer and more robust policies by incorporating human expert data and representation learning into DRL.

B. Reinforcement Learning With Human Demonstrations

Utilizing human expert prior knowledge or human guidance in DRL to enhance sample efficiency and reduce dependence on reward engineering is considered a promising approach. One method involves pre-training with imitation (supervised) learning, initializing the policy using expert demonstrations to achieve a certain level of performance before applying reinforcement learning to optimize the policy further guided by the reward function [14], [15]. However, this approach often performs poorly, especially for maximum entropy reinforcement learning [3], [16], which encourages policy stochasticity for

initial exploration. Another method integrates expert demonstrations into the experience replay buffer of off-policy DRL algorithms, sampling experiences from both expert demonstrations and agent interactions to update the policy [17], [18]. Yet, this requires labeling each state-action pair with rewards to conform to the transition tuple format used in reinforcement learning, which can be challenging due to the unavailability of expert rewards.

Huang et al. [19] proposed a method that first obtains an expert policy network using behavioral cloning (BC), then guides learning by constraining the KL divergence between the agent's policy and the cloned expert policy. However, this approach is limited by BC's out-of-distribution generalization challenges and sets BC as a prerequisite operation, affecting policy adaptability and algorithm flexibility.

In contrast, our study employs an expert policy discriminator updated synchronously with DRL to construct policy constraints, allowing for dynamic expansion of the expert dataset, thereby overcoming the aforementioned challenges.

C. Contrastive Learning

Contrastive learning learns data representations through similarity and dissimilarity constraints, akin to performing anchor-based dictionary lookups. In instance discrimination, different augmented versions of the same instance form positive pairs, while different instances constitute negative pairs [20]. Overly iterative updates of neural networks can lead to overfitting. Research has shown that learning from the differences between positive and negative samples during training can effectively prevent overfitting [21]. Other studies have aimed at obtaining better representation information to enhance DRL decision-making capabilities. The work in [22] introduced discriminative contrastive learning, which effectively captures environmental dynamics relevant to decision-making without additional computational complexity. Reference [8] highlighted the importance of applying contrastive learning in reinforcement learning, particularly by optimizing the InfoNCE objective to extract highly relevant environmental information for decision-making.

Inspired by the applications of contrastive learning across various domains, our study adopts contrastive learning as an auxiliary task for identifying expert policies, enabling our method to continuously and accurately recognize expert policies.

III. PRELIMINARIES

A. Scene Description and Vehicle Control

1) *Scene Description*: We utilized the SMARTS simulation platform [25] to construct two typical urban driving scenarios—unprotected left turns and ramp merging—to evaluate the efficacy of the proposed algorithm in urban autonomous driving. As shown in Fig. 2-(a), the unprotected left turn requires the agent to cross a four-lane, two-way road and complete a left turn into the main road without traffic signal guidance, reaching the rightmost lane. Fig. 2-(c) displays the first-person perspective of the ego vehicle, facilitating observation and analysis by human experts. The ramp merging

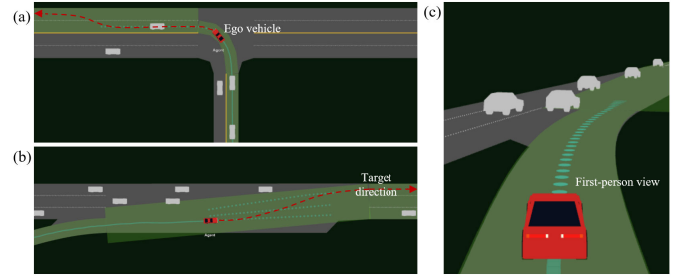


Fig. 2. Urban driving scenarios designed within the SMARTS platform. (a) Unprotected left turn. (b) Merge scenario at a slip road. (c) First-person view of expert demonstrations.

scenario depicted in Fig. 2-(b) simulates the process of safely merging onto the main road from a ramp and entering the leftmost lane. The green areas in Fig. 2 indicate feasible routes, providing intuitive navigation references. These two scenarios thoroughly test the adaptability and decision-making capabilities of the algorithm.

To simulate diverse driver behaviors, we assigned different driving behavior model parameters (such as speed distribution, operational errors, impatience levels, and willingness to cooperate) to the environmental vehicles based on the method described in [19]. This ensures that each traffic flow consists of a mix of various types of participants. During training, 30 different traffic flow configurations were used, with initial conditions (such as vehicle positions and speeds) randomly generated for each scenario. In the testing phase, an additional 40 traffic flows were employed to validate the generalization performance of the algorithm.

2) *Vehicle Control*: In this study, the autonomous vehicle acts as an agent whose behavior decisions are dominated by the DRL algorithm. The high-level control commands generated by the DRL algorithm are converted into specific low-level control signals through a closed-loop Proportional-Integral-Derivative (PID) controller. The lateral position control of the vehicle is achieved using the following formulas:

$$v_{lat} = -K_{lat}\Delta_{lat}, \quad \Delta\psi_r = \arcsin\left(\frac{v_{lat}}{v}\right) \quad (1)$$

where Δ_{lat} is the lateral distance of the vehicle relative to the lane centerline, v_{lat} is the lateral speed command, and $\Delta\psi_r$ is the heading change applied due to the lateral speed command.

The heading control of the vehicle is performed using the following equations:

$$\psi_r = \psi_L + \Delta\psi_r, \quad \dot{\psi}_r = K_\psi(\psi_r - \psi), \quad \delta = \arcsin\left(\frac{l}{2v}\dot{\psi}_r\right) \quad (2)$$

Here, ψ_L represents the lane heading, ψ_r is the target heading that follows the lane heading and position, $\dot{\psi}_r$ is the yaw rate command, δ is the front wheel steering angle control signal, l is the wheelbase of the vehicle, and K_{lat} and K_ψ are the control gains for position and heading control, respectively.

The vehicle's motion is determined by the bicycle kinematic model described as follows:

$$\begin{aligned} \dot{x} &= v \cos(\psi + \beta_{angle}), \quad \dot{y} = v \sin(\psi + \beta_{angle}), \\ \dot{v} &= a, \quad \dot{\psi} = \frac{v}{l} \sin(\beta_{angle}), \quad \beta_{angle} = \arctan\left(\frac{1}{2} \tan(\delta)\right) \end{aligned} \quad (3)$$

where (x, y) are the position coordinates of the vehicle, v is the forward speed, ψ is the heading, a is the acceleration command, and β_{angle} is the sideslip angle at the center of mass.

B. Maximum Entropy Reinforcement Learning

The problem of making a sequence of decisions for autonomous vehicles in complex and dynamic environments can be modeled as a Markov Decision Process (MDP) [23]. An MDP is defined by the tuple (S, A, P, R, γ) , where S is the state space with $s \in S$, A is the action space with $a \in A$, P is the state transition distribution denoted as $P(s' | s, a)$ representing the probability of transitioning from state s to state s' upon executing action a , $R(s, a)$ is the reward function, and $\gamma \in (0, 1]$ is the discount factor ensuring diminishing weight of future rewards, balancing immediate and long-term benefits.

MaxEnt RL aims to derive an optimal agent policy π within systems adhering to the MDP framework, balancing between maximizing the expected discounted total reward and promoting exploration by maximizing the policy's entropy. The policy $\pi(a_t | s_t)$ defines the probability distribution over actions given state s_t , aiming not only to maximize the cumulative rewards but also to maximize the policy entropy to encourage a wider exploration of the action space:

$$\underset{\pi}{\text{maximize}} \quad \mathbb{E}_{\tau} \left[\sum_{t=0}^T \gamma^t R(s_t, a_t) + \alpha \mathcal{H}(\pi(\cdot | s_t)) \right], \quad (4)$$

where τ denotes trajectories generated under $a_t \sim \pi(a_t | s_t)$ and $s_{t+1} \sim P(s_{t+1} | s_t, a_t)$, T is the maximum time step of trajectory τ , α is the temperature coefficient ($\alpha > 0$), and $\mathcal{H}(\pi(\cdot | s_t)) = -\sum_{a_t} \pi(a_t | s_t) \log \pi(a_t | s_t)$ represents the entropy of the policy. This formulation emphasizes that the goal is to find a policy that maximizes both the expected return and the entropy, leading to more exploratory behaviors.

C. MDP Formulation

1) *State Space*: As shown in Fig. 4, with the rise of end-to-end autonomous driving technology [1], [2], we pay special attention to the decision-making and control tasks for autonomous driving that use Bird's Eye View (BEV) images as state inputs. Since a single image contains little information about object velocities, which can be estimated using previous images, it lacks the Markov property. Therefore, we stack the most recent three BEV images to form the state input $s_t = (o_t, o_{t-1}, o_{t-2})$, where $s_t \in \mathbb{R}^{C \times H \times W}$, with C , H , and W representing the number of channels, height, and width of the stacked BEV images, respectively, as shown in Fig. 2.

2) *Action Space*: The action command is a tuple containing target speed and lane change instructions $a_t = (v_{\text{target}}, l)$, where $a_t \in \mathbb{R}^2$. Both v_{target} and l are normalized to the interval $[-1, 1]$. A value of $l \geq 0.5$ indicates a lane change to the right, while $l \leq -0.5$ indicates a lane change to the left. Finally, we obtain the final control motion commands through a PID controller to interact with the environment.

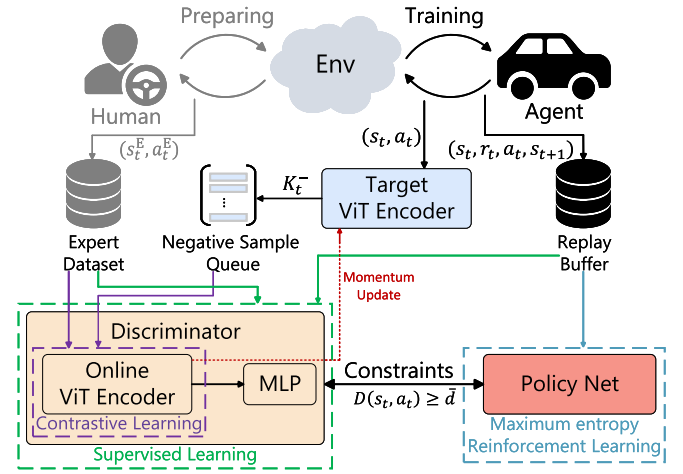


Fig. 3. Conceptual framework of Contrastive Expert-Guided Maximum Entropy Reinforcement Learning (CE-MERL). In the figure, the “Discriminator” refers to the “Expert Policy Discriminator” that we have proposed.

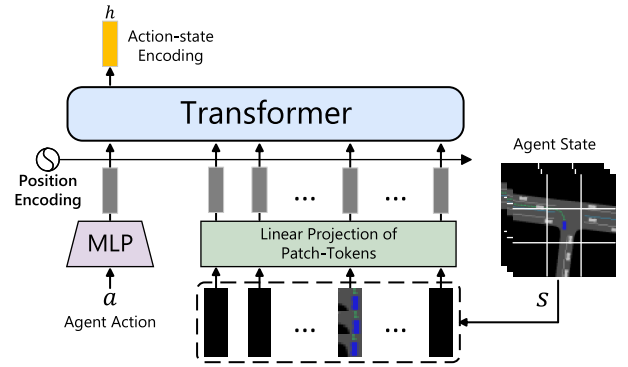


Fig. 4. ViT Encoder: Adopting a state-action joint encoder based on ViT, the deep integration of scene representation and action instruction information is achieved by utilizing the action encoding as classification query tokens. This approach guides the scene representation to focus on features relevant to the action instructions in subsequent learning processes. It is worth noting that there exist a large number of redundant feature areas irrelevant to decision-making within the scene state composed of BEV images.

3) *Reward Function*: Our method, CE-MERL, employs a sparse reward function, allowing the agent to receive a reward signal only at the end of an episode:

$$R(s, a) = r_{\text{collision}} + r_{\text{goal}} \quad (5)$$

Here, $r_{\text{collision}} = -1$ when the vehicle collides, goes off-road, or takes a wrong direction; otherwise, $r_{\text{collision}} = 0$. Similarly, $r_{\text{goal}} = 1$ when the vehicle successfully reaches its goal; otherwise, $r_{\text{goal}} = 0$.

IV. METHOD

Fig. 3 illustrates the conceptual framework of CE-MERL. This framework initially gathers data from human experts, followed by iterative optimization of both the expert policy discriminator and the policy network during the training phase. An enhanced version of ViT is employed as the feature encoder for the discriminator, with a contrastive learning auxiliary task introduced to boost its training effectiveness. On this foundation, behavioral constraints are implemented

using the discriminator's output, concurrently conducting maximum entropy reinforcement learning to optimize the policy network.

In this section, we first introduce the key component of the expert policy discriminator in the CE-MERL framework—the state-action joint feature encoder based on ViT. We then describe the method for updating the expert policy discriminator with a contrastive learning auxiliary task. Finally, we explain how to use this discriminator to construct constraints and integrate them into the MaxEnt RL framework to guide policy optimization.

A. State-Action Joint Feature Encoder Based on ViT

To enable our expert policy discriminator to identify the implicit policy information within states and actions, it is necessary to promote high coupling between the encoded information of states and actions. Therefore, we employ a state-action joint encoder based on ViT as the core component for feature extraction.

As shown in Fig. 4, given an input image $x \in \mathbb{R}^{C \times H \times W}$, where C , H , and W represent the number of channels, height, and width of the image, respectively, ViT first segments the state visual image s into several patches $x = (x_1, x_2, \dots, x_n)$, where $x_n \in \mathbb{R}^{N \times (p^2 c)}$ and $N = HW/p^2$, and maps these patches to linear embeddings:

$$z_0 = [x_0; LP(x_1); LP(x_2); \dots; LP(x_n)] + pos \quad (6)$$

where LP denotes linear projection, $pos \in \mathbb{R}^{(N+1) \times D}$ represents positional embedding encoding, and $x_0 \in \mathbb{R}^{1 \times D}$ is a learnable query token. Subsequently, z_0 passes through l layers of Transformer encoders, each containing a Multi-Head Self-Attention (MSA) mechanism and a Feed-Forward Neural Network (FFN) [12].

In our work, as shown in Fig. 4, we take the scene state s_t as the input image and use a Multi-Layer Perceptron (MLP) to map the action a_t to a query token $c \in \mathbb{R}^{1 \times D}$, where $c = MLP(a_t)$. Thus, the state-action joint encoder based on ViT can be represented as:

$$h = ViT(s, a; \varphi) \quad (7)$$

where φ denotes the model parameters. This approach facilitates the coupling of scene representation with action instruction information and focuses the scene representation primarily on features related to action instructions during subsequent learning processes.

B. Expert Policy Discriminator

The expert policy discriminator is a neural network model designed to handle binary classification tasks, determining whether input experience samples conform to the distribution of expert policies. It consists of the aforementioned ViT encoder and an MLP classifier, formulated as follows:

$$d = D(s, a; \varphi, \omega) = \text{Sigmoid}(MLP(ViT(s, a; \varphi); \omega)) \quad (8)$$

Here, D represents the expert policy discriminator model, ω are the parameters of the MLP, and $d \in \mathbb{R}^1$ is the

prediction outcome. This model judges if a given experience tuple originates from the expert dataset or if, given state s , the input action a comes from an expert policy. Specifically, (s_t^E, a_t^E) denotes tuples from the expert dataset, while (s_t, a_t) come from the replay buffer in off-policy DRL. The loss function for the expert policy discriminator is defined as:

$$L_D(\varphi, \omega) = -\log D(s_t^E, a_t^E; \varphi, \omega) - \log(1 - D(s_t, a_t; \varphi, \omega)) \quad (9)$$

Typically, the scale of datasets provided by human experts is significantly smaller compared to the amount of data accumulated through agent interactions. This asymmetry in data volume leads to imbalanced distributions between the two types of data, which can cause the expert policy discriminator to overfit. Specifically, the discriminator may become overly sensitive to non-critical features of states or actions in the input scenes rather than focusing on key attributes that define expert decision patterns. Consequently, such overfitting hinders the discriminator's ability to accurately identify and reproduce expert strategies.

To address this issue, we incorporate a contrastive learning auxiliary task into the learning process of the expert policy discriminator, as illustrated in Fig. 5. We use the ViT encoder as both a query encoder and a key-value encoder, with the parameters of the key-value encoder φ' updated via momentum (Polyak averaging), i.e., after each epoch, $\varphi' = \lambda\varphi + (1 - \lambda)\varphi'$. Given two expert experience tuples (s_i^E, a_i^E) and (s_j^E, a_j^E) where $i \neq j$, they undergo different data augmentations to obtain $s_i^{E'}$ and $s_j^{E'}$. Our data augmentation includes random horizontal flipping, random rotation, random cropping around the ego vehicle, and random geometric transformations. To add noise that does not affect the decision, for $a_t = (v_{\text{target}}, l)$, we introduce a random noise in the range of $[-1, 1]$ to v_{target} . For l , if $l \leq -0.5$, the added noise ensures $l < -0.5$; if $l \geq 0.5$, the noise ensures $l > 0.5$. This approach maintains action boundaries and avoids unintended transitions between discrete states.

We define the query vector $q = ViT(s_i^{E'}, a_i^{E'}; \varphi)$, and the positive sample $k^+ = ViT(s_j^{E'}, a_j^{E'}; \varphi)$. The right side of Fig. 5 illustrates the construction of the negative sample queue. In the DRL loop, each time step experience tuple of the agent is encoded by the Target ViT Encoder, $k_t^- = ViT(s_t', a_t; \varphi')$, where s_t' represents s_t after one data augmentation, and then stored in a queue of length N_k . Further, our negative samples consist of N_k negative sample vectors from the queue and $k_{N_k+1}^- = ViT(s_t^{E'}, a_t; \varphi')$, where $a_t \sim \pi(\cdot | s_t^E)$. Therefore, the contrastive learning loss term is:

$$L_{NCE}(\varphi) = -\log \frac{\exp(q \cdot k^+ / \tau_l)}{\exp(q \cdot k^+ / \tau_l) + \sum_{i=1}^{N_k+1} \exp(q \cdot k_i^- / \tau_l)}, \quad (10)$$

where τ_l is the temperature hyperparameter. Thus, the overall loss function for the expert policy discriminator with the contrastive learning auxiliary task is:

$$L_{CD}(\varphi, \omega) = L_{NCE}(\varphi) + L_D(\varphi, \omega). \quad (11)$$

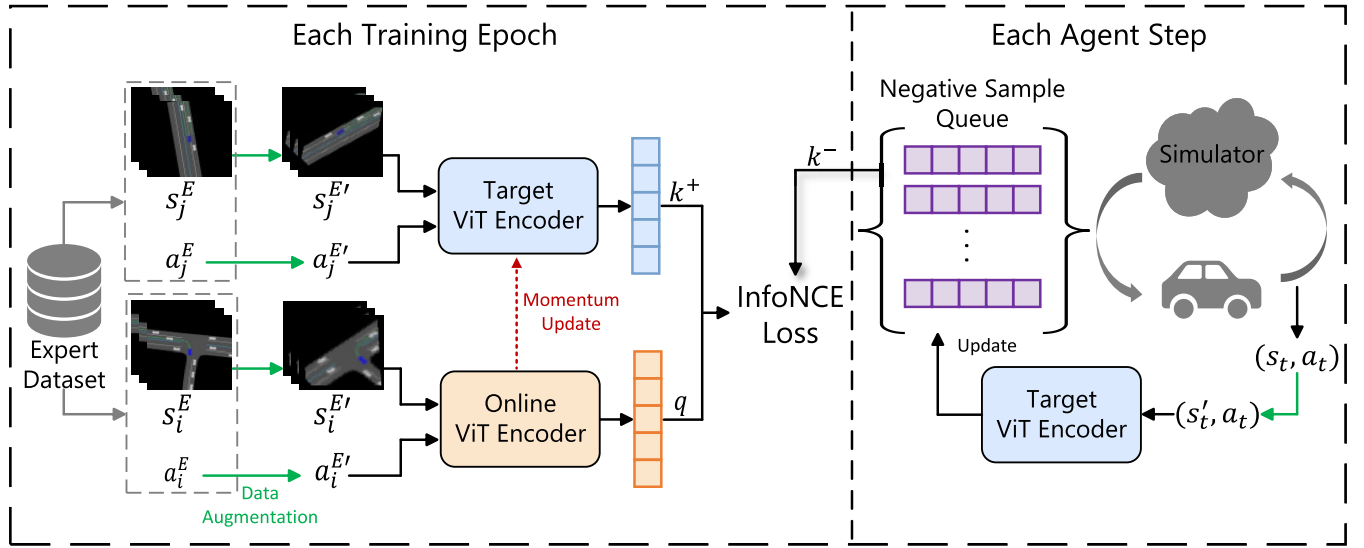


Fig. 5. Contrastive learning: The left panel illustrates the auxiliary task employed during each epoch of training to update the state-action joint feature encoder (Online ViT Encoder) within the expert policy discriminator. The right panel explains the construction of the negative sample queue: at each step of interaction between the agent and the simulator, the state-action pairs are encoded by the Target ViT Encoder for updating the negative sample queue. The Target ViT Encoder is obtained through a momentum update from the Online ViT Encoder at the end of each epoch.

In the expert policy discriminator design, our aim is to identify policy info within state-action pairs and verify their origin from an expert dataset, especially under imbalanced data distribution. Utilizing a modified ViT-based joint state-action encoder with a contrastive learning auxiliary task, we maximize mutual information for expert state-action embeddings while enhancing distinctions between expert and agent data features. This improves intra-class compactness and inter-class separability, mitigating overfitting due to asymmetric data distribution. Consequently, the discriminator accurately extracts policy details and distinguishes expert from non-expert data.

C. Contrastive Expert-Guided Maximum Entropy Reinforcement Learning

Following the auto-adjusted temperature coefficient approach of the SAC algorithm [3], we reformulate the optimization objective described in Eq. 4 by incorporating the entropy term as a constraint. Subsequently, we integrate the cost constraints defined by the expert policy discriminator D into the optimization objective, thereby constructing the CE-MERL framework. This process underscores the importance of simultaneously enhancing exploration (through entropy constraints) and ensuring that the learned behaviors align with those demonstrated by experts (through cost constraints), aiming to achieve more reliable and safer policy learning. Thus, the optimization objective for CE-MERL is:

$$\begin{aligned} \max_{\pi} \quad & \mathbb{E}_{\tau} \left[\sum_{t=0}^T \gamma^t R(s_t, a_t) \right] \\ \text{s.t.} \quad & \begin{cases} \mathbb{E}_{(s_t, a_t) \sim \tau} [-\log \pi(a_t | s_t)] \geq \mathcal{H} \\ \mathbb{E}_{(s_t, a_t) \sim \tau} [D(s_t, a_t; \varphi, \omega)] \geq \bar{d}, \end{cases} \end{aligned} \quad (12)$$

where $\bar{d}$ represents the minimum expected value output by the discriminator model.

For the agent learning step, we propose a modified version of the Soft Actor-Critic (SAC) algorithm [3]. We add the output of the expert policy discriminator as an additional reward signal to the reward function, allowing CE-MERL to operate without relying on complex shaped rewards. Therefore, the state value function Q is recursively defined as:

$$\begin{aligned} Q(s_t, a_t) = & R(s_t, a_t) + \gamma \mathbb{E}_{(s_{t+1}, a_{t+1}) \sim \tau} [Q(s_{t+1}, a_{t+1}) \\ & - \alpha \log \pi(a_{t+1} | s_{t+1}) + \beta D(s_{t+1}, a_{t+1}; \varphi, \omega)], \end{aligned} \quad (13)$$

where α and β are two temperature coefficients used to balance exploration, imitation, and exploitation. Given a replay buffer $B = \{(s_t, a_t, r_t, s_{t+1})\}_{t=1}^N$, for the update and sampling of B , we adopted the Prioritized Experience Replay technique [24]. we use two Q-function networks $Q(s_t, a_t; \phi_1)$ and $Q(s_t, a_t; \phi_2)$, along with two target Q-function networks with parameters $\bar{\phi}_1$ and $\bar{\phi}_2$, to minimize the Mean Squared Bellman Error (MSBE) in order to mitigate overestimation issues:

$$\begin{aligned} L_Q(\phi_i) = & \mathbb{E}_{\substack{(s_t, a_t, r_t, s_{t+1}) \sim B \\ a_{t+1} \sim \pi(\cdot | s_{t+1}; \theta)}} \left[\frac{1}{2} \left(Q(s_t, a_t; \phi_i) \right. \right. \\ & - \left(r_t + \gamma \min_{i=1,2} Q(s_{t+1}, a_{t+1}; \bar{\phi}_i) \right. \\ & \left. \left. - \alpha \log \pi(a_{t+1} | s_{t+1}) + \beta D(s_{t+1}, a_{t+1}; \varphi, \omega) \right) \right)^2 \right]. \end{aligned} \quad (14)$$

After each epoch, $\bar{\phi}_i = \eta_i \phi_i + (1 - \eta_i) \bar{\phi}_i$, for $i = 1, 2$. For the policy network $\pi(\cdot | s_t; \theta)$ with parameters θ , it maximizes the discounted cumulative reward in Eq. (12) by

maximizing the output of the value network Q . Due to the non-convexity of the constraints (attributable on one hand to the neural network characteristics of the discriminator network D and the policy network π , and on the other hand to the adversarial generative training process of D), it is challenging to strictly satisfy the constraints in continuous space. To address this, CE-MERL transforms the original problem into a relaxed Lagrangian dual form:

$$\begin{aligned} \max_{\alpha} \max_{\beta} \min_{\pi} \mathbb{E}_{\substack{s_t \sim B \\ a_t \sim \pi(\cdot|s_t; \theta)}} \left[- \min_{i=1,2} Q(s_{t+1}, a_{t+1}; \phi_i) \right. \\ \left. + \alpha \left(\mathcal{H} + \log \pi(a_t|s_t; \theta) \right) + \beta \left(\bar{d} - D(s_t, a_t; \varphi, \omega) \right) \right]. \end{aligned} \quad (15)$$

We then adopt the dual gradient descent method, alternating gradient updates over the parameters θ and the Lagrange multipliers α and β to optimize the unconstrained objective function. The policy network utilizes gradient signals to guide the policy towards the feasible region without strictly enforcing constraint satisfaction. Thus, the corrected loss function for the policy network is defined as:

$$\begin{aligned} L_{\pi}(\theta) = \mathbb{E}_{s_t \sim B} \left[\mathbb{E}_{a_t \sim \pi(\cdot|s_t; \theta)} \left[- \min_{i=1,2} Q(s_{t+1}, a_{t+1}; \phi_i) \right. \right. \\ \left. \left. + \alpha \left(\mathcal{H} + \log \pi(a_t|s_t; \theta) \right) \right. \right. \\ \left. \left. + \beta \left(\bar{d} - D(s_t, a_t; \varphi, \omega) \right) \right] \right]. \end{aligned} \quad (16)$$

The parameters α and β are updated according to the following loss functions, with the constraint that $\alpha, \beta \geq 0$:

$$L_E(\alpha) = \mathbb{E}_{s_t \sim B} \left[\mathbb{E}_{a_t \sim \pi(\cdot|s_t; \theta)} \left[- \alpha \left(\mathcal{H} + \log \pi(a_t|s_t; \theta) \right) \right] \right], \quad (17)$$

$$L_C(\beta) = \mathbb{E}_{s_t \sim B} \left[\mathbb{E}_{a_t \sim \pi(\cdot|s_t; \theta)} \left[- \beta \left(\bar{d} - D(s_t, a_t; \varphi, \omega) \right) \right] \right]. \quad (18)$$

In summary, CE-MERL maintains the agent's exploration capabilities while implicitly constraining its policy to be close to expert policies, leading to more reasonable exploration actions and safer, more robust behavior policies. Additionally, the discriminator's output serves as an extra reward signal, reducing the reliance of DRL on complex reward designs. Algorithm 1 summarizes the key steps of the algorithm.

V. EXPERIMENTAL SETUP

A. Human Expert Demonstrations Data

We required human experts to complete driving tasks safely and without violations in all randomly generated traffic scenarios provided by the simulator, except for timeout failures caused by traffic congestion. For the two designated tasks of unprotected left turns and ramp merging, most human drivers were able to meet this requirement after a short training period involving less than 10 trials. To mitigate bias due to individual driving styles, we recruited four human experts to perform the designed urban driving tasks within the simulator, collecting

Algorithm 1 CE-MERL

Require: expert dataset D^E , model parameters $\varphi, \omega, \theta, \phi_1, \phi_2$, batch size b , replay buffer B of size N^B , negative queue Q^N of length N^{Queue} for state-action encodings, learning rates l_D, l_{π}, l_Q, l_T .

Ensure: Initialize empty Q^N , empty B and temperature parameters α, β .

```

1: Warm-up Phase:
2: for each warm-up step do
3:   Interact with the environment using random actions.
4:   Collect experience  $(s_t, a_t, r_t, s_{t+1})$  and store it in  $B$ .
5:   Encode augmented state-action pairs into  $Q^N$ :
      $k_t^- \leftarrow \text{ViT}(s'_t, a_t; \varphi')$ ;
6:   Add  $k_t^-$  to  $Q^N$ .
7: end for
8: Training:
9: for each epoch do
10:  for each agent step do
11:    Collect experience:  $(s_t, a_t, r_t, s_{t+1}) \rightarrow B$ .
12:    Encode augmented state-action pairs into  $Q^N$ .
13:  end for
14:  for each gradient step do
15:    Sample batches  $\{(s_t, a_t)\}_1^b$  and  $\{(s_t^E, a_t^E)\}_1^b$  from
       $B$  and  $D^E$ .
16:     $\{(s_t^{E'}, a_t^{E'})\}_1^b \leftarrow \text{Augment}(\{(s_t^E, a_t^E)\}_1^b)$ ;
17:    Construct query and positive samples using
       $\{(s_t^{E'}, a_t^{E'})\}_1^b$ .
18:     $NegSet \leftarrow Q^N$ ;
19:    for  $i = 1$  to  $b$  do
20:       $a_t \leftarrow \pi(\cdot|s_t^E; \theta)$ ;
21:       $k_t^- \leftarrow \text{ViT}(s_t^E, a_t; \varphi')$ ;
22:      Add  $k_t^-$  to  $NegSet$ .
23:    end for
24:    Update  $\varphi, \omega$  via  $L_D(\varphi, \omega)$ .
25:    Soft update  $\varphi' = \lambda \varphi + (1 - \lambda) \varphi'$ .
26:    Update  $\phi_i, i \in \{1, 2\}$  via  $L_Q(\phi_i)$ .
27:    Soft update  $\bar{\phi}_i = \eta_i \phi_i + (1 - \eta_i) \bar{\phi}_i, i = 1, 2$ .
28:    Update  $\theta$  via  $L_{\pi}(\theta)$ .
29:    Update  $\alpha, \beta$  via  $L_E(\alpha), L_C(\beta)$ .
30:  end for
31: end for

```

their operations as expert demonstration data. The human experts observed the driving environment from a first-person perspective, as illustrated in Fig. 2(c), and provided their actions through keyboard inputs for four discrete actions. Two discrete actions were used for longitudinal control: acceleration (target speed increased by 2 m/s) and deceleration (target speed decreased by 2 m/s). Similarly, two discrete actions were employed for lateral movement: lane change to the left and lane change to the right, with the default operation being to maintain the current lane. Human experts could simultaneously execute both longitudinal and lateral control actions but were not required to provide input at every time step. We collected one set of expert driving data for each of the unprotected left turn and ramp merging tasks, with each dataset consisting of 40 successful trajectories.

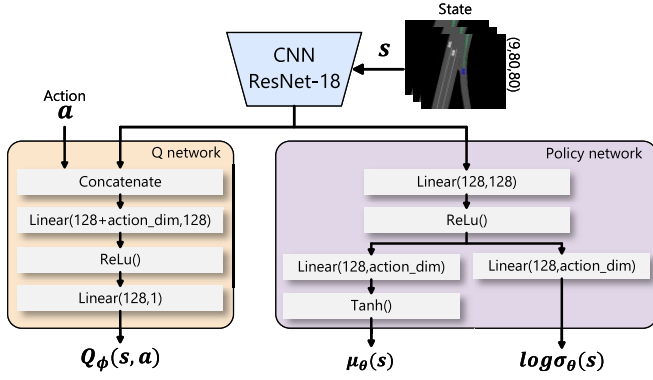


Fig. 6. The network architectures of the key components in DRL, the Critic (Q-network) and the Actor (Policy network), which share a CNN feature encoder.

B. Implementation Details

In our study, we adopted a ViT-based encoder architecture as the feature extractor for the expert policy discriminator. Specifically, our ViT encoder configuration was as follows: we used image patches of size as the basic units of input. To balance model complexity and computational efficiency, we selected a two-layer Transformer structure, each layer containing two attention heads. The dimension of each attention head was set to 128, matching the embedding dimension of the model to ensure efficient information transmission. Furthermore, to enhance the expressive power of the model, we appended a MLP with a 512-dimensional output after each Transformer layer for nonlinear feature mapping. The network architectures of the Actor and Critic components in DRL are illustrated in Fig. 6. The action distribution, involving two continuous actions for longitudinal and lateral motion control, was modeled as a multivariate Gaussian distribution, using only the mean outputs as actions during evaluation and testing. The policy network generated parameters for the Gaussian distribution, while the Q-value network output a single value for each. Unless otherwise specified in Fig. 6, all layers utilized the ReLU activation function. All networks were constructed using the Pytorch framework and trained with the Adam optimizer.

The hyperparameters used in the proposed method are summarized in Table I. A portion of these hyperparameters were adopted from the default settings in the SAC hyperparameter repository, which have been validated across multiple prior works and facilitate comparisons with other baseline algorithms. For other hyperparameters such as the “Size of negative sample encoding queue N^{Queue} ”, we followed the recommended configurations from the CoDy [8] study. The “Discriminator learning rate l_D ” and “Discriminator’s Minimum Expected Output d ” were determined through grid search to identify their optimal values.

C. Comparison Baselines

To comprehensively evaluate the effectiveness of CE-MERL, we selected two categories of state-of-the-art DRL algorithms for comparison. One category consists of DRL algorithms that do not rely on human expert demonstrations:

Symbol	Meaning	Value
N^B	Size of replay buffer	2×10^4
N^{Queue}	Size of negative sample encoding queue	6144
b	Training batch size	256
N^{warm}	Warm-up steps	10^3
N^{train}	Total training steps	10^5
l_π, l_Q	DRL learning rate	0.0002
l_D	Discriminator learning rate	0.001
l_T	Temperature parameters learning rate	0.002
γ	Discount rate	0.98
λ	Target ViT encoder momentum coefficient	0.1
η_1, η_2	Target Q net momentum coefficient	0.02
$\mathcal{H}$	Target entropy	-2
$\bar{d}$	Discriminator’s Minimum Expected Output	0.5

- **SAC [3]**: A leading off-policy reinforcement learning algorithm, we utilized a version that can automatically adjust the temperature parameter.
- **CoDy [8]**: This method leverages contrastive learning to enhance the Markov property of learned embeddings and combines it with SAC, thereby improving the model’s sample efficiency and generalization.
- **PPO [26]**: A widely recognized on-policy reinforcement learning algorithm that uses clipped policy gradients to ensure the new policy does not diverge too far from the old one, stabilizing policy training.

However, based on our experiments and previous studies, achieving at least acceptable performance solely through sparse rewards without prior knowledge from expert demonstrations is almost impossible for DRL baseline algorithms. Therefore, to improve the success rate of DRL baselines without expert demonstrations, we employed shaping rewards for these algorithms:

$$R'(s, a) = 0.01 r_{\text{speed}} + r_{\text{collision}} + r_{\text{goal}} - 0.001 r_{\text{distance}} \quad (19)$$

where r_{speed} represents the current speed of the ego vehicle, and r_{distance} denotes the distance to the goal point.

The other category includes advanced DRL algorithms or imitation learning methods that incorporate human expert demonstrations:

- **EPRL [19]**: This approach introduces the KL divergence between the expert policy distribution estimated by BC and the agent’s policy as a regularization term, reducing sample complexity and decreasing dependency on reward design.
- **BC+DDPG [14], [15]**: By pretraining the policy network with BC and subsequently applying the DDPG [27] algorithm, this method provides an efficient way to integrate expert prior knowledge into DRL.
- **GAIL [28]**: An advanced imitation learning (IL) method based on generative adversarial learning paradigms, often compared with DRL due to its similar learning process.

Note that $R(s, a)$ (5) is used for all DRL baseline algorithms incorporating expert demonstrations and our proposed CE-MERL method, while $R'(s, a)$ (19) is applied to all DRL

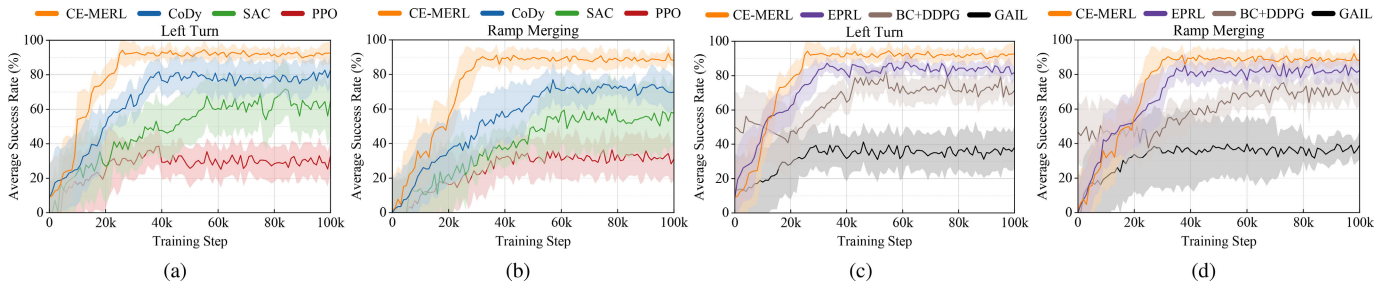


Fig. 7. This figure illustrates the convergence curves of task success rates for our proposed CE-MERL and various baseline algorithms across different tasks. Subfigures (a) and (b) compare CE-MERL against baseline algorithms without human expert demonstrations, while subfigures (c) and (d) present comparisons with baselines that incorporate human expert demonstrations; Subfigures (a) and (c) correspond to the unprotected left-turn task, whereas subfigures (b) and (d) relate to the merging onto a slip road task. Each task was performed for 20 episodes in the evaluation environment to compute the average success rate, and the experiments were repeated 10 times using different random seeds. The shaded areas in the figures indicate the range of one standard deviation.

TABLE II

COMPARISON OF TASK SUCCESS RATES (%) BETWEEN CE-MERL AND VARIOUS DRL BASELINE ALGORITHMS IN TWO URBAN DRIVING SCENARIOS: UNPROTECTED LEFT TURN AND RAMP MERGING

	CE-MERL	CoDy	SAC	PPO	EPRL	BC+DDPG	GAIL
Left Turn	96	84	72	36	94	81	41
Ramp Merging	92	81	65	38	90	78	40

The success rate is obtained by running the optimal model, acquired during training, for 100 episodes.

baseline algorithms without expert demonstrations. For baselines combining human demonstration data and our proposed CE-MERL, the same expert dataset was used.

VI. EXPERIMENTAL RESULTS AND DISCUSSION

A. Comparison and Analysis of CE-MERL With Baseline Algorithms

This section introduces and analyzes the performance of our method in two tasks: unprotected left-turn and ramp merging, compared with baseline algorithms with and without expert priors. Fig. 7 shows the convergence curves of success rates for CE-MERL and various baseline algorithms, while Table II presents the test success rates of the optimal models obtained during training for CE-MERL and the baseline algorithms.

As shown in Fig. 7a, Fig. 7b and Table II, CE-MERL demonstrates higher sample efficiency and success rates compared to DRL baseline algorithms without human expert demonstrations, even when using only sparse rewards. The CoDy and SAC algorithms, relying solely on the maximum entropy framework for random policy search, result in excessive unnecessary exploration and unstable learning processes, as evidenced by the larger error ranges. CoDy enhances the Markovian property of state-action pairs through contrastive learning, achieving a more stable learning process and improving the sample efficiency and success rates of SAC. For the on-policy PPO algorithm, the high-dimensional state and action spaces lead to entrapment in local optima. CE-MERL, in addition to the maximum entropy constraint, introduces an extra constraint that the state-action samples generated by the agent's interaction with the environment should conform to the human expert policy distribution. In this way, we not only ensure the agent's exploration capability but also bring its behavioral policy closer to the expert policy, better balancing exploration and exploitation, thereby enhancing success rates and sample efficiency.

As illustrated in Fig. 7c, Fig. 7d and Table II, CE-MERL also exhibits superior performance compared to advanced baseline algorithms with human expert demonstrations (none of which employ reward shaping techniques). The EPRL algorithm introduces the KL divergence between expert and agent policies as a constraint, an approach similar to ours. However, EPRL relies on behavior cloning (BC) to approximate the expert policy, leading to significant estimation errors in KL divergence due to distribution shift. In contrast, our method utilizes a ViT encoder combined with contrastive learning techniques to achieve a more accurate estimation of the expert policy. For the BC+DDPG method, although the policy network is initialized with BC, this results in incorrect guidance from the Critic network to the policy network during the early stages of training. Moreover, the combination of BC and DDPG further restricts the agent's exploration capability, thereby affecting the overall performance of the algorithm. The CE-MERL method we propose adds policy constraints within the MaxEnt RL framework to mitigate the negative impact of expert priors on the agent's exploration capability. Both EPRL and BC+DDPG separate the learning of expert and agent policies into two distinct and sequential subtasks, limiting flexibility. Our method integrates the learning of expert and agent policies into a unified task framework, allowing for dynamic updates and adjustments of the expert dataset, enhancing the system's flexibility and adaptability. Finally, the GAIL algorithm, due to its reliance on extensive and high-quality datasets, fails to learn acceptable policies under our dataset and environment settings.

B. The Impact of Reward Shaping and Expert Priors on DRL

Fig. 8 illustrates the average speed and acceleration of our proposed CE-MERL algorithm and various baseline algorithms when completing an unprotected left-turn mission. The unprotected left-turn, due to its more complex and delicate

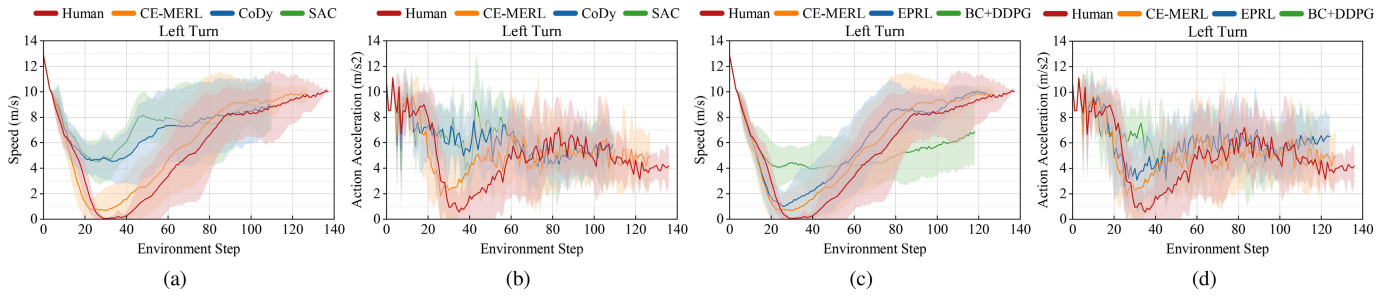


Fig. 8. The figure presents the average speed and acceleration of the optimal policy models obtained by our proposed CE-MERL algorithm and various baseline algorithms during training, as well as those of human experts, in the unprotected left-turn task (the averages are computed based on 40 successful trajectories aligned to the shortest trajectory). Specifically, subfigures (a) and (b) display the average speed and acceleration per step for human experts, the CE-MERL algorithm, and baseline algorithms without expert demonstration datasets; subfigures (c) and (d) show the average speed and acceleration per step for human experts, the CE-MERL algorithm, and baseline algorithms with expert demonstration datasets.

TABLE III

COMPARISON OF THE AVERAGE DURATION (IN SECONDS) FOR CE-MERL AND VARIOUS DRL BASELINE ALGORITHMS TO COMPLETE TASKS IN TWO URBAN DRIVING SCENARIOS: UNPROTECTED LEFT TURN AND RAMP MERGING

	CE-MERL	CoDy	SAC	PPO	EPRL	BC+DDPG	GAIL
Left Turn	15.18 ± 1.14	12.70 ± 0.63	11.42 ± 0.76	25.73 ± 0.95	15.30 ± 0.84	13.91 ± 2.46	26.24 ± 4.15
Ramp Merging	22.48 ± 2.24	18.32 ± 3.22	17.60 ± 2.70	27.16 ± 1.88	22.80 ± 1.09	19.23 ± 3.01	38.03 ± 5.27

The average duration is calculated based on the optimal model obtained during training, which successfully completes 100 episodes.

human-machine interaction process and denser mixed traffic flow, is more capable of revealing the differences among various strategies. Therefore, we only present the average speed and acceleration for the unprotected left-turn task, while the average accelerations of all algorithms tend to be similar in the ramp merging task. Table III shows the average time taken by strategies learned using different methods to complete different tasks. Additionally, since PPO and GAIL did not learn acceptable policies, they are not discussed in this section.

Impact of Reward Shaping on DRL: The CoDy and SAC algorithms complete tasks in less time because the speed rewards in reward shaping prompt them to adopt extremely aggressive and unsafe driving strategies, such as accelerating at intersections to force oncoming vehicles to stop and create a passage gap. Theoretically, to enable such algorithms to learn safe and robust strategies, the key lies in designing a reward function that can accurately depict the complex human-machine interaction process. However, this remains a significant unresolved challenge in the current research field. It is worth noting that the design of such a reward function is essentially equivalent to the design of a rule-based autonomous driving decision control algorithm. On one hand, past research has shown that rule-based autonomous driving decision-making control algorithms generally perform poorly; on the other hand, this also goes against the original intention of using DRL or deep learning techniques to construct an autonomous driving decision control algorithm.

Impact of Expert Priors on DRL: The strategy learned by EPRL is closer to that of human experts because it separates the imitation of expert strategies as a preliminary task, avoiding overfitting to the expert strategies. Consequently, during the DRL phase, its approximated expert strategy positively guides the agent, bringing the agent's strategy closer to the expert strategy. However, due to the out-of-distribution

generalization problem of BC, EPRL is limited in its final performance. In contrast, our proposed CE-MERL can more accurately depict the expert strategy, thereby achieving superior performance in the final results. The policy network of the BC+DDPG method has already learned a certain level of strategy before the start of DRL and can provide successful trajectory experiences for the agent during the DRL warm-up period, enabling it to successfully converge relying solely on sparse rewards. However, during the DRL process, due to the lack of continuous guidance from the expert strategy, the algorithm fails to learn a safer and more robust strategy similar to that of human experts.

Within the CE-MERL framework, the expert strategy discriminator, by leveraging a ViT encoder and contrastive learning, maximizes the mutual information of expert data, effectively reducing the neural network's excessive attention to non-strategy-related scene features. This method not only enhances the recognition accuracy of the expert strategy but also prevents overfitting, continuously exerting a positive influence on the agent's strategy during the DRL phase. Furthermore, [30] points out that, compared to explicit models such as policy networks, implicit models like expert strategy discriminators and Q-networks are more effective in identifying and capturing expert strategies. Under the influence of these factors, our proposed CE-MERL outperforms various baseline algorithms and learns safer and more robust human-like strategies. If the expert prior ablation is applied to the CE-MERL method, the algorithm will degenerate into a form equivalent to the SAC algorithm.

C. Analysis of the Effectiveness of ViT State-Action Encoder and Contrastive Learning

In this section, we analyze and verify the effectiveness of the ViT state-action encoder and the auxiliary task of con-

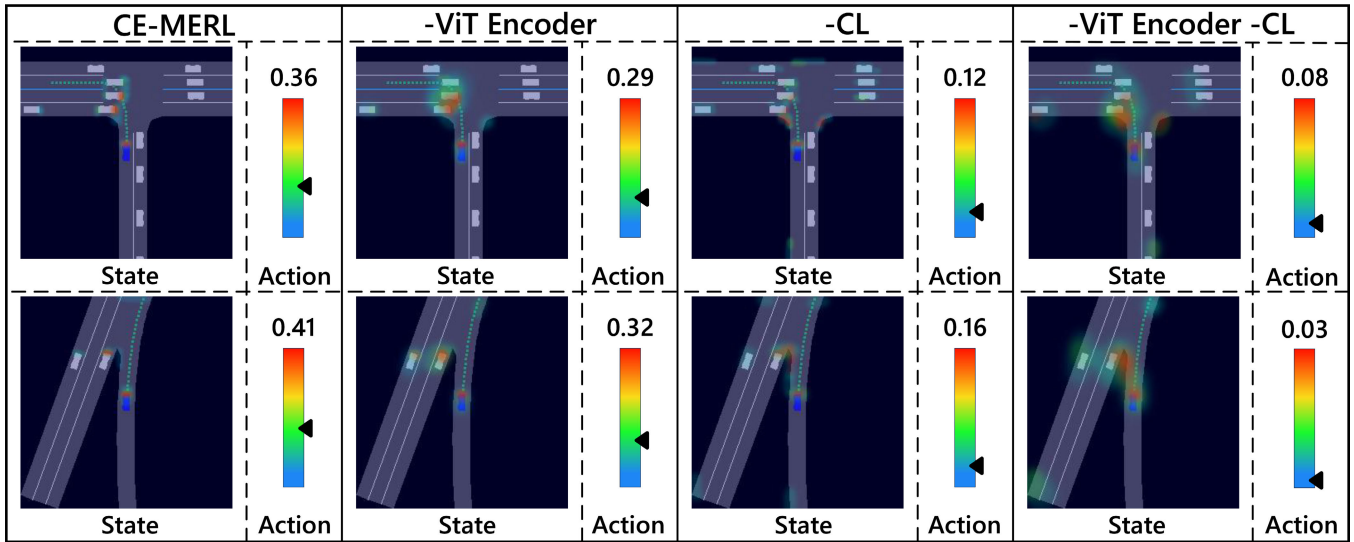


Fig. 9. The expert policy discriminator of CE-MERL takes multimodal inputs, including stacked BEV images and action command vectors. This figure not only illustrates the attention distribution of the discriminator on BEV images but also presents the extent of the contribution of action commands to the decision-making. The first row samples a set of state-action pairs from the unprotected left turn expert dataset; the second row samples a set of state-action pairs from the ramp merging expert dataset. Each column corresponds to an expert policy discriminator with a different configuration. In each subplot, the left side shows the effect of the state input image (composed of three stacked RGB images, with only the top layer shown for simplified visualization) modified by the attention mask; the right side illustrates the contribution of action inputs to the decision-making, where the numerical values represent the importance scores of the computed action vectors. The intensity of attention is indicated by a color gradient from blue (low) to red (high).

trastive learning in the expert strategy discriminator through attention visualization techniques and ablation experiments. Figures 9, 10, and 11 correspond to three experiments, respectively. Variants that remove the ViT state-action joint encoder within the expert strategy discriminator are denoted as “-ViT Encoder”, while configurations that omit the contrastive learning loss term are marked as “-CL”. All “-ViT Encoder” variants of the expert discriminator adopt the same network structure as the Critic component, as shown in Fig. 6.

In the expert strategy discriminator attention flow visualization experiment of Fig. 9, the “-ViT Encoder” and “-ViT Encoder -CL” adopted the CNN architecture, with their attention weights calculated and displayed using the Grad-CAM method [29]. The discriminators in CE-MERL and those labeled with “-CL” are based on the ViT architecture; for this reason, we employed a variant of the Grad-CAM adapted for ViT to compute the attention distribution and the importance score of action commands [12]. The experimental results show that the discriminator model with the ViT state-action encoder can more accurately focus on relevant features, effectively filtering out state information unrelated to policy representation. Moreover, by embedding the action input as a query token, the ViT structure ensures that the importance of action information in the decision-making process is not overlooked or lost. The introduction of contrastive loss as an auxiliary task further optimizes the attention mechanism, focusing it on features crucial to policy expression, such as the state features of the ego vehicle and surrounding social vehicles. In contrast, when the contrastive learning auxiliary task is removed, due to the multi-step iterative nature of DRL and the imbalance in the distribution of expert data and agent interaction data, the model tends to overfocus on non-critical

factors such as road structure or self-localization, leading to overfitting.

As shown in Fig. 10, after removing the contrastive loss task from the expert strategy discriminator, its ability to correctly discriminate significantly decreases, failing to accurately determine whether the input experience conforms to the expert strategy. Removing the ViT state-action encoder results in a more dispersed output score distribution of the discriminator. In CE-MERL, the expert policy discriminator D retains a strong ability to identify expert policies under the adversarial generative framework for the following reasons: Due to the coupling effect of the hyperparameter $\bar{d}$ and the temperature coefficient β in $L_{\pi}(\theta)$, the discriminator’s gradient signals may be diluted, leading to lower optimization efficiency compared to the explicit adversarial loss $L_{CD}(\varphi, \omega)$. This phenomenon aligns with theoretical analyses of early convergence of discriminators in GANs. Additionally, introducing contrastive learning auxiliary tasks enhances the abstraction capability of state-action encoding, alleviating mode collapse issues in adversarial training. The discriminator’s continuous and accurate identification of expert policies in CE-MERL is crucial for enforcing constraints that implicitly limit the distance between the agent’s policy and the expert policy. Furthermore, our experiments show that the optimized policy model does not strictly satisfy the cost constraint defined by the expert policy discriminator D ; that is, not all state-action pairs generated by the policy network are identified by the discriminator as expert policies with a probability greater than $\bar{d}$. For the adversarial generative framework, this indicates that CE-MERL does not suffer from mode collapse. In the context of DRL, CE-MERL employs a relaxed Lagrangian dual form to optimize the policy, ensuring it does not solely rely on

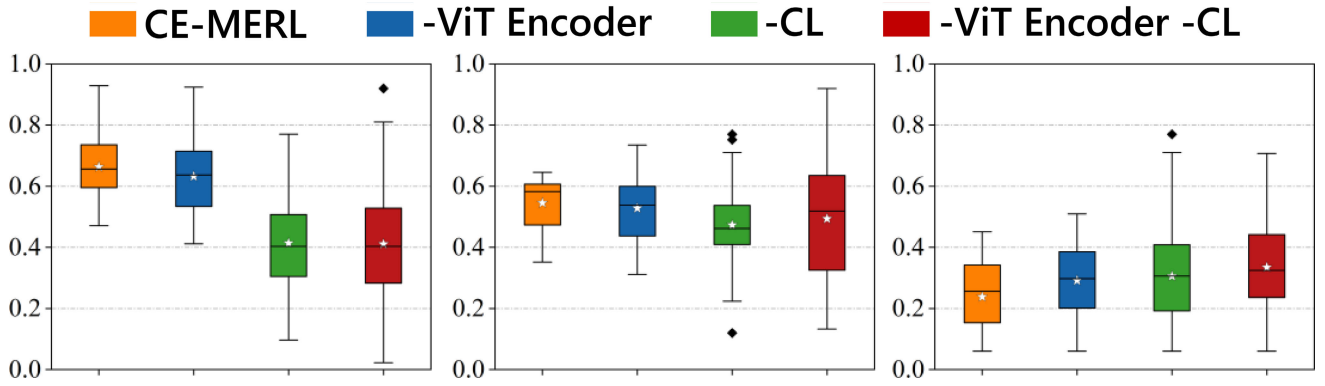


Fig. 10. This figure presents a boxplot of the expert policy discriminator's predictions for 128 state-action samples from various sources after CE-MERL has undergone 100,000 iterations. The vertical axis displays the output values of the expert policy discriminator, indicating the probability that the input state-action samples belong to the expert policy. The data for the left boxplot originates from the expert dataset; the middle boxplot reflects samples drawn from the replay buffer of CE-MERL; the right boxplot combines three types of samples: 64 interaction samples generated by a random policy, 32 states selected from the expert dataset paired with randomly generated actions, and another 32 samples composed of states from the CE-MERL replay buffer combined with random actions.

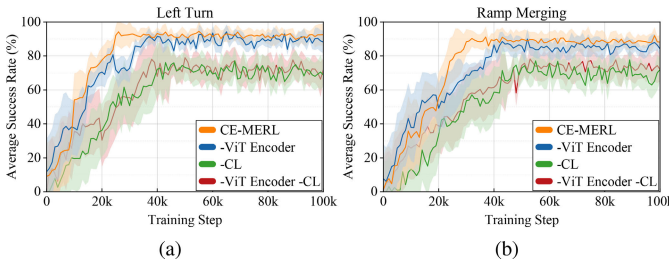


Fig. 11. The figure illustrates a performance comparison in the CE-MERL framework, where the complete configuration of CE-MERL is contrasted with variants that have various components of the expert policy discriminator removed, across two urban driving tasks: unprotected left turn and ramp merging.

imitation but instead contracts into a feasible region balancing exploitation, exploration, and imitation.

Fig. 11 shows that, in the absence of a contrastive loss term, the performance of CE-MERL remains similar regardless of whether a ViT state-action joint feature encoder is adopted. This similarity in performance primarily benefits from the foundational contribution of the pre-trained ResNet network. However, once the contrastive loss term is integrated into the expert policy discriminator, the advantages of the ViT encoder become significantly apparent, highlighting its core value in enhancing model performance. This experiment further confirms the importance of the ViT encoder and the auxiliary task of contrastive learning to CE-MERL, a conclusion consistent with the analysis results of the previous two experiments.

The experimental results in this section collectively demonstrate that by employing the ViT encoder for deep coupling of state-action pairs and combining it with the auxiliary task of contrastive learning, overfitting is effectively prevented, and the accuracy of the expert strategy discriminator in identifying expert strategies is significantly improved. This enhanced recognition capability is key to CE-MERL's rapid convergence and outstanding performance.

VII. CONCLUSION

In this study, we discuss an innovative approach, the CE-MERL framework, for integrating human experts' prior knowledge into deep reinforcement learning (DRL). This

method introduces a consistency measure between surrogate and expert policies as an additional cost constraint within the maximum entropy reinforcement learning framework. Simulated experiments on unprotected left turns and ramp merging tasks demonstrate that CE-MERL exhibits significant advantages in sample efficiency, policy safety, and robustness compared to various state-of-the-art baseline algorithms. Furthermore, we analyze potential issues of reward shaping techniques in autonomous driving tasks and verify the superiority of leveraging human experts' prior knowledge as an alternative solution. CE-MERL not only circumvents the need for complex reward design but also promotes autonomous driving decision-making closer to human behavioral patterns. Therefore, CE-MERL provides a novel solution not only to the field of autonomous driving but also contributes to the application of autonomous driving algorithms in mixed human-machine real-world environments. On a broader scale, CE-MERL can be applied to multiple domains such as drone delivery and intelligent traffic management. However, within the current framework, CE-MERL relies on simulator environments and their generated ground-truth BEV images, making it currently unsuitable for direct application in the real world. Future research will continue to explore the possibility of using real-world datasets with CE-MERL, such as incorporating advanced BEV generation algorithms into our framework to enable its application in real-world autonomous driving tasks.

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